Concurrency and Distributed Systems

Week 6 – Part 3: Parallel Streams Anti-Example

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Part 3 Roadmap

1. The Temptation
2. Primitive vs Boxed
3. Common Pool
4. Slowdown Causes
5. Exercise 3

The Temptation

Common mistake

Adding `parallel()` to a stream does not mean the program becomes faster.

The stream pipeline must still split work, schedule tasks, execute lambdas, combine results, and compete for common-pool workers.

Sequential Primitive Sum

```
long total = IntStream.rangeClosed(1, size)
        .asLongStream()
        .sum();
```

This avoids boxing and uses a primitive stream specialized for numeric work.

Deliberately Bad Parallel Version

```java
long total = IntStream.rangeClosed(1, size)
        .boxed()
        .parallel()
        .mapToLong(Integer::longValue)
        .sum();
```

The parallel version computes the same result but adds avoidable costs.

Four Slowdown Causes

1. **Boxing overhead:** primitive ints become `Integer` objects.
2. **Lambda overhead:** pipeline stages add call and allocation costs.
3. **Thread contention:** common-pool workers are shared.
4. **Splitting overhead:** the stream must partition and combine work.

Common Pool Contention

- Parallel streams normally use the common `ForkJoinPool`.
- Other code in the JVM may use the same pool.
- Blocking or long-running common-pool work can delay unrelated stream work.

Operational risk

Hidden shared pools make performance harder to reason about.

When Parallel Streams Can Be Acceptable

- Large input.
- Independent elements.
- CPU-bound operation per element.
- Cheap splitting and combining.
- No shared mutable state.
- Measured win on the target environment.

Demo06

- `sequentialPrimitiveSum(size)` is the baseline.
- `parallelBoxedSum(size)` is the anti-example.
- `slowdownCauses()` names the explanation students must give.

Exercise 3 Contract

- Sequential and parallel versions must compute the same value.
- Diagnosis must include exactly the four required causes.
- The answer must not claim that parallel streams are always faster.

Exit Check

1. Why does boxing hurt numeric stream performance?
2. What does the common pool make implicit?
3. When would you accept a parallel stream in production?